

RESEARCH ARTICLE

A Mobile Health Approach for Glaucoma Detection

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ABSTRACT Glaucoma is a progressive eye disease that ranks as the second leading cause of blindness and the foremost cause of irreversible blindness globally. According to the National Institutes of Health's National Eye Institute (NIH-NEI), glaucoma accounts for 3 million cases of irreversible blindness and low vision in the United States. Many individuals with glaucoma may exhibit no symptoms, making early detection essential for preventing blindness and preserving vision. This paper presents a novel smartphone-based method for the early detection and prediction of glaucoma. Our approach employs an innovative CIE LAB color-based enhancement technique for preprocessing, as well as a 3D stereo depth mapping method to extract the optic disc and optic cup from retinal fundus images. We calculate the areas of the optic disc and optic cup to determine the cup-to-disc ratios. For classification, we utilized a support vector machine (SVM). Our proposed method demonstrated a remarkable ability to differentiate glaucomatous eyes from healthy ones, achieving a sensitivity of 96.6%, specificity of 98.2%, and an overall accuracy of 97.1%.

INDEX TERMS Glaucoma, support vector machine (SVM), cup-to-disc ratio, image processing.

I. INTRODUCTION

Glaucoma is a progressive ocular condition characterized by the deterioration of the optic nerves. It is recognized as the leading cause of irreversible blindness globally, and its prevalence continues to rise. Currently, it is estimated that glaucoma affects 91 million individuals worldwide [1], [3]. Approximately three million individuals in the United States have received a diagnosis of glaucoma. Within this population, 2.7 million individuals aged 40 and older are impacted by the most prevalent form of the disease, known as open-angle glaucoma. The risk factors associated with glaucoma include increasing age, African American ethnicity, diabetes, reduced central corneal thickness, elevated intraocular pressure (IOP), and a familial history of the condition. Glaucoma is often asymptomatic in its early stages, leading to gradual vision loss that may go unnoticed until significant optic nerve damage has occurred. This is particularly true for open-angle glaucoma, the most common type. However, in advanced

stages or in cases of acute angle-closure glaucoma, symptoms such as severe eye pain, blurred vision, and halos around lights may occur [4], [5]. Due to these factors, detection methods capable of diagnosing glaucoma in its early stages or predicting its onset have garnered significant attention in recent years, making a substantial impact on public health. In today's landscape, computer vision and machine learning have emerged as mainstream techniques for the accurate and timely detection of retinal diseases, including macular degeneration (MD), glaucoma, and diabetic retinopathy (DR) [7]. Key features for diagnosing retinal diseases include the optic nerve disc, retinal blood vessels, and the fovea. The optic disc (Fig. 1) is a bright, circular structure comprised of retinal nerves and blood vessels, making it a crucial element in diagnosing various conditions, particularly glaucoma and foveal disorders. The optic nerve, composed of numerous axons from retinal ganglion cells, is responsible for transmitting visual information from the retina to the brain. Glaucoma leads to progressive damage to these axons, resulting in irreversible vision loss. Glaucoma primarily affects the peripheral vision in its early stages, progressing towards

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central vision loss as the disease advances. Unlike macular diseases, which directly impact the fovea, glaucoma results from progressive optic nerve damage that leads to characteristic visual field defects [8]. The extraction of optic disc features in retinal images is crucial for the development of automated glaucoma screening systems. Recent methods for glaucoma detection utilize shape, appearance, and intensity features to pinpoint the optic disc location. For instance, Sinthanayothin et al. in [9] and [10] proposed a technique that detects the optic disc by focusing on the highest variability in adjacent pixel intensities. However, this method tends to struggle with retinal images that contain large areas of bright lesions, as it may misidentify these areas as optic discs. Therefore, the shape of the optic disc is an important feature for effectively distinguishing it from other regions of the retina. In [11], Park et al. utilized brightness and shape features for the detection and extraction of optic discs from retinal images. This method employs the Hough transform to delineate high-intensity optic disc contours. A significant limitation of this approach is that bright artifacts resulting from conditions such as diabetic retinopathy, as well as noisy brighter spots, may also be erroneously classified as optic discs, thereby compromising detection performance. Furthermore, the features of optic discs exhibit variability in size, color, and shape, which complicates their detection using the aforementioned method [12]. To enhance the accuracy of optic disc detection, template matching techniques have been proposed in [8], [9], and [10]. However, the presence of bright spots and noise in retinal imagery poses challenges in identifying a template that accurately corresponds to optic discs. Recent investigations have shifted focus towards the extraction of retinal vascular features for optic disc detection. In [13], the location of the optic disc was determined through the intersection of retinal blood vessels. Kumar et al. [14] employed geometrical features that link the optic disc to nearby blood vessels for its localization. Chen et al. [15] introduced the vessels' directional matched filter (VDMF) for the extraction of the optic disc's location, while Russo et al. [16] utilized a multiscale VDMF combined with a Gaussian filter to identify the optic disc center, defined as the pixel with minimal difference from its neighboring pixels. Although the use of vascular tree structures could potentially enhance the accuracy of optic disc extraction, the segmentation of vascular trees remains a labor-intensive and complex procedure. In addition to these techniques, machine learning methodologies for optic disc localization have been explored. In [17], Orlando et al. implemented a cascade classifier for optic disc localization, utilizing Haar features for training the classifier and subsequent extraction. Nonetheless, this method is both time-consuming and computationally intensive. In another study, Raghavendra et al. [18] integrated active shape detection methods with principal component analysis (PCA) to locate the optic disc by calculating the minimum distance between the retinal image and its projection within the disk space. A drawback of this methodology is the difficulty in

determining an appropriate shape model capable of accommodating the diverse shapes of optic discs. Glaucoma is classified into primary open-angle glaucoma (POAG) and angle-closure glaucoma (ACG) based on the anatomy of the anterior chamber angle between the iris and the cornea. While elevated intraocular pressure (IOP) is a major risk factor for glaucoma, it is not the sole basis for classification, as some forms of glaucoma, such as normal-tension glaucoma, occur despite normal IOP levels.

Open-angle glaucoma (OAG) is a progressive ocular disease that affects the optic nerve, leading to a deterioration of peripheral visual function and resulting in an excavated appearance of the optic disc as observed through ophthalmoscopy. The precise etiology of OAG remains unclear; however, intraocular pressure (IOP) is recognized as a significant risk factor. Other contributing risk factors include a family history of glaucoma, race, advanced age (60 years and older), history of eye injuries, diabetes, and corneal thickness. The high costs associated with existing glaucoma screening methods pose a substantial barrier to identifying individuals who are at elevated risk of blindness due to glaucoma. Research has been conducted utilizing fundus and optical coherence tomography (OCT) imaging techniques for the detection of glaucoma. Automated detection systems that analyze fundus and OCT images have been proposed. These studies typically focus on identifying and measuring features of the optic nerve. The predominant approach to glaucoma detection via retinal fundus images involves quantifying the dimensions of the optic disc, optic rim, and optic cup to calculate the cup-to-disc ratio (CDR) or the optic disc-to-optic rim ratio (ODR). A CDR greater than 0.5 indicates a diagnosis of glaucoma, whereas a ratio below 0.5 is considered indicative of a normal eye. In this paper, we propose an innovative glaucoma prediction and detection system that leverages smartphone technology. This system integrates identified risk factors with fundus retinal images to assess and predict an individual's current glaucoma status or their risk of developing glaucoma in the future. The established method for glaucoma detection involves evaluating the size and shape of the optic disc, optic cup, and optic rim, while localizing the nerve head's position. Research into glaucoma detection through fundus retinal imaging has included measurements of the optic nerve's size and shape, as well as the extraction of rim-to-disc (RDR) and cup-to-disc ratios (CDR).

Traditional optic disc detection methods face several challenges, including variations in optic disc shape, interference from bright lesions and blood vessels, inconsistent image quality, and difficulties in segmenting the optic cup accurately. Many existing approaches rely on template matching, thresholding, or edge detection, which often fail in cases of irregularly shaped optic discs or low-contrast images. To overcome these challenges, our method integrates CIE LAB color space conversion for enhanced contrast, disparity-based depth mapping for precise cup segmentation, and an SVM classifier that combines cup-to-disc ratio (CDR)

features with depth cues. These advancements improve robustness against anatomical variations, reduce errors in cup segmentation, and enhance glaucoma detection accuracy in real-world settings.

The utilization of smartphones to capture and process images in this study is advantageous compared to traditional cameras. Modern smartphones are equipped with advanced processing capabilities that facilitate image enhancement, processing, and automated analysis for the identification of ocular diseases. For instance, preliminary image preprocessing, such as cropping and resizing, can be effectively performed on smartphones, while machine learning techniques can be implemented using the generated images. Additionally, smartphone-based diagnostic systems benefit from ubiquitous internet connectivity, enabling integration with electronic patient records and incorporation into patient management systems.

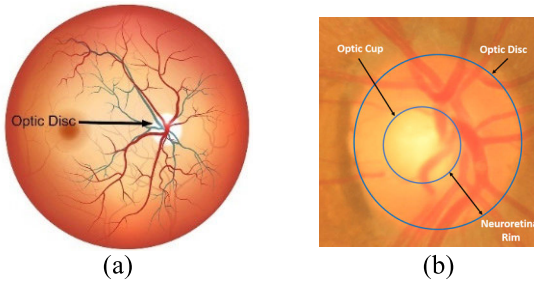


FIGURE 1. Illustration of the optic disc and optic cup on a retina: (a) A demonstration of the optic disc, and (b) an annotated sample of a fundus retina image highlighting the optic disc and optic cup.

II. MATERIALS AND METHODS

In this paper, retinal fundus images were obtained from the G1020 dataset, an extensive collection containing 1,020 publicly accessible images, including 724 healthy and 296 glaucoma cases. This dataset is aimed at aiding glaucoma diagnosis and includes labeled images along with segmentation data for both the optic disc and optic cup [26]. The G1020 dataset presents a unique challenge in classifying healthy and glaucomatous eyes due to its diversity and real-world imaging conditions. It consists of 1,020 high-resolution fundus images (724 healthy, 296 glaucomatous), with images acquired using a conventional fundus camera under clinical conditions. Unlike curated datasets, G1020 retains images with varying lighting, noise, and optic disc positioning, making segmentation and classification more complex. The dataset includes cases of varying glaucoma severity, with cup-to-disc ratio (CDR) measurements provided for each image. However, the ground truth diagnosis was not based solely on CDR but was clinically verified using optic nerve head appearance, retinal nerve fiber layer (RNFL) defects, intraocular pressure (IOP) measurements, and ISNT rule compliance. A subset of 250 healthy and 250 glaucomatous cases were selected to ensure age comparability and diversity in disease severity, optic disc morphology, and image quality. These factors

collectively make G1020 a robust benchmark for evaluating glaucoma detection models under real-world clinical conditions. Our proposed method starts with a preprocessing step. In the preprocessing step, the images are analyzed to reduce noise and enhance the color contrast. In the next step, the images are converted to CIE LAB color space. Next, the images are segmented, and the region of interest, which in this case is the optic nerve cup and disc, is extracted. In the next step, the disc and cup areas are calculated. Finally, the support vector machine (SVM) classifier was used to extract the optic disc and optic cup, and the results are presented. The steps of the proposed method are shown in Fig. 2.

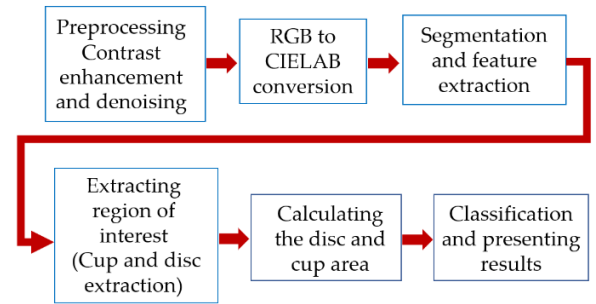


FIGURE 2. Flowchart of our proposed method: Data acquisition and preprocessing. Data were acquired from eye dataset using retinal fundus images of normal eyes and eyes with glaucoma.

A. PROCESSING AND FEATURE EXTRACTION

Given a retina image, segmentation is first applied to locate all foreground retina pixels and extract the cup and disc area. To extract the optic disc and optic cup areas from the pre-processed retinal fundus image, we employed a novel image processing technique that distinguishes the cup and disc areas based on their color features. (CIE LAB) color space was adopted to distinguish the designated areas using their color features.

To convert RGB to CIELAB color space, we first convert RGB color space to CIEXYZ using the following equation:

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} 0.4124 & 0.3576 & 0.1805 \\ 0.2126 & 0.7152 & 0.0722 \\ 0.0193 & 0.1192 & 0.9505 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix} \quad (1)$$

Next, CIEXYZ to CIELAB color space conversion is done using the following equations:

$$L^* = 116f\left(\frac{Y}{Y_0}\right) - 16, \quad (2)$$

$$a^* = 500 \left[f\left(\frac{X}{X_0}\right) - f\left(\frac{Y}{Y_0}\right) \right], \quad (3)$$

$$b^* = 200 \left[f\left(\frac{Y}{Y_0}\right) - f\left(\frac{Z}{Z_0}\right) \right], \quad (4)$$

where

$$f(x) = \begin{cases} x^{1/3} & (x > 0.008856), \\ 7.787x + \frac{16}{116} & (x \leq 0.008856), \end{cases} \quad (5)$$

After converting the RGB image to CIE LAB format, our proposed method extracted the color features of the optic disc and optic cup to determine cup to disc and cup to rim ratio.

B. CUP TO DISC RATIO (CDC)

A standard method for detecting glaucoma is measuring the cup-to-disc ratio, but accurately determining the cup and disc area is challenging and prone to errors. The cup-to-disc ratio (CDR) is commonly used as an indicator of glaucomatous damage, but there is significant natural variation in CDR among individuals and across different age groups. While a CDR greater than 0.5 has been associated with an increased risk of glaucoma, it is not an absolute diagnostic criterion. Some healthy individuals may naturally have larger CDRs, while some glaucomatous eyes may have CDRs below this threshold. Therefore, glaucoma diagnosis cannot rely on CDR alone and must be confirmed through additional clinical evaluations, including optic nerve head appearance, intraocular pressure (IOP) measurements, visual field testing, and retinal nerve fiber layer (RNFL) analysis. Most studies approximate these areas as circles or ellipses, which may not reflect their true shape. To overcome this, we propose a novel method that directly counts all cup and disc pixels, ensuring accurate area measurement regardless of shape or position. The cup-to-disc ratio is then calculated using these precise measurements, making the method independent of image size.

C. SEGMENTATION

The segmentation step is required to extract the optic nerve head region (region of interest) and to distinguish the optic cup area from the optic disc area. For the segmentation step, we adopted the color features from the CIE LAB color space and segmented different regions of the retina based on their color. The following table shows the allocated regions from different colors in the retinal images:

TABLE 1. CIE LAB threshold values for segmentation of retina, optic disc, and optic cup.

CIE LAB values	L	A	B
Retina	$13.3 < L < 58.6$	$-1.2 < A < 37.8$	$24.4 < B < 59.3$
Optic Disc	$70.7 < L < 73.8$	$24.9 < A < 32.2$	$75.6 < B < 78.2$
Optic Cup	$78.35 < L < 85.2$	$0.21 < A < 14.1$	$80.4 < B < 85.8$

The results of the segmentation step are shown in the following image. In Fig. 3, the optic disc is shown with green color, the optic cup is shown with blue color and the retina background is shown in its original color. Our proposed method is shown in Fig. 3b and the automatic optic disc and cup method used in [9] is shown in Fig. 3c for comparison.

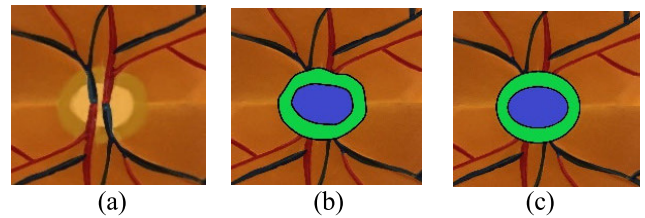


FIGURE 3. Segmentation of the cup and disc area using our proposed method: (a) original image, (b) cup and disc marked using our proposed method, and (c) cup and disc marked using an automated cup and disc detection system.

D. DISPARITY AND DEPTH MAP

Disparity refers to the distance between two corresponding points in the left and right images of a stereo pair. The cup-to-disc and cup-to-rim ratio has drawbacks when it comes to detecting Glaucoma, especially in the early stage of the disease. Hence, ophthalmologists use alternative methods and diagnostic tools such as Optical Coherence Tomography (OCT) to diagnose Glaucoma in the early stage of the disease [27]. Using OCT, ophthalmologists can map the retina in 3D images and study the retina and optic nerve head layer by layer [28]. To diagnose glaucoma in an accurate, reliable way, our proposed method combines the CDC method with 3D stereo depth mapping to determine the optic cup shape and diagnose glaucoma from both the optic nerve head color and the magnitude of cupping.

The proposed smartphone-based 2D-to-3D imaging system enhances early glaucoma detection by incorporating a novel depth mapping approach that improves upon traditional 2D fundus imaging. The system captures stereo fundus images from slightly different angles and generates a disparity map using the Sum of Absolute Differences (SAD) algorithm. By leveraging CIE LAB color space conversion for enhanced contrast and segmentation, it extracts depth information of the optic nerve head to identify glaucomatous cupping more accurately. The integration of machine learning, specifically an SVM classifier trained on both CDR features and depth cues, enables improved classification of healthy versus glaucomatous eyes. Unlike conventional fundus photography, which lacks depth perception, or costly optical coherence tomography (OCT), our smartphone-based system provides an affordable, portable, and scalable solution for early glaucoma screening, particularly in low-resource settings.

Depth mapping plays a crucial role in enhancing glaucoma detection by enabling precise quantification of optic cup excavation, improving cup-to-disc ratio (CDR) measurements, and detecting neuroretinal rim thinning. Our system utilizes stereo disparity mapping, employing the Sum of Absolute Differences (SAD) algorithm to generate a depth map from two slightly different fundus images. The optic disc and cup are segmented using adaptive thresholding and Canny edge detection in the CIE LAB color space.

Depth-based features, including cup depth profile, neuroretinal rim thickness, and a depth-adjusted CDR, are extracted and used in a Support Vector Machine (SVM) classifier. Additionally, bicubic interpolation is applied for smooth 3D surface reconstruction of the optic nerve head. By incorporating depth-based analysis, our approach provides a more accurate and early detection mechanism for glaucoma compared to conventional 2D imaging methods.

Our stereo imaging system was evaluated using images from an existing dataset and tested on multiple smartphone models with both dual- and single-camera configurations. For dual-camera phones (e.g., iPhone 13 Pro, Samsung Galaxy S22 Ultra), stereo images were captured natively using built-in disparity mapping. For single-camera devices (e.g., Google Pixel 6, Xiaomi Mi 11), stereo imaging was achieved by slightly tilting the phone during capture or using an external fundus adapter. A traditional monocular fundus camera with controlled lateral movement was also used to validate stereo acquisition in clinical settings. The system showed strong performance across devices, confirming its adaptability for consumer and clinical use. The disparity map we acquired from 2 different angles is used to create a disparity map. The left and right images developed a stereo image (Fig. 5). By adopting the following equation, we created a disparity map using the sum of absolute difference algorithm (SAD) [29].

$$SAD(x, y, d) = \sum_{(x,y) \in w} |I_l(x, y) - I_r(x - d, y)| \quad (6)$$

In this equation, (x, y, d) is the coordination for the disparity map, (x, y) is the pixel of interest coordinates, and d is the depth (or disparity) value. Usually, in the matching process, the right image (I_r) is used as a target and the left image (I_l) represents the reference image. The differences are summed over the aggregated support window w to generate a simple metric of block similarity known as a disparity map.

The Sum of Absolute Differences (SAD) algorithm is adopted for computing the matching cost. The SAD algorithm is capable of functioning in real time, which makes it appropriate to be operated in a smartphone. The results of disparity mapping for a sample healthy and diseased retina is shown in Fig. 4.

Classification For the classification step, we adopted the support vector machine (SVM) algorithm by utilizing the sigmoid kernel function [30]. The images were divided into testing and training sets for the classification step. The training set was later divided into training and validation subsets and the testing set was left out specifically for the testing procedure. Thus, we adopt SVM to classify our samples (x_i) into healthy ($y_i = 1$) and diseased ($y_i = -1$). These two statuses are classified with the hyperplane ($f(x)$) which is defined as:

$$f(x) = w^T \Phi(x) + b \quad (7)$$

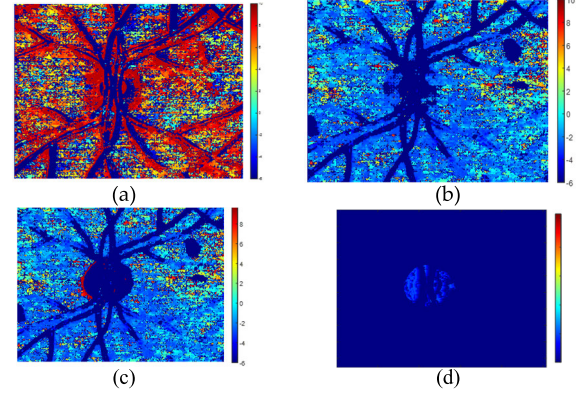


FIGURE 4. Disparity map of the healthy and diseased retina using stereo imaging: a) disparity map of the retinal vein system, b) disparity map of a healthy retina, c) disparity map of the diseased retina, and d) disparity map of the optic nerve cup (cupping) of a diseased eye.

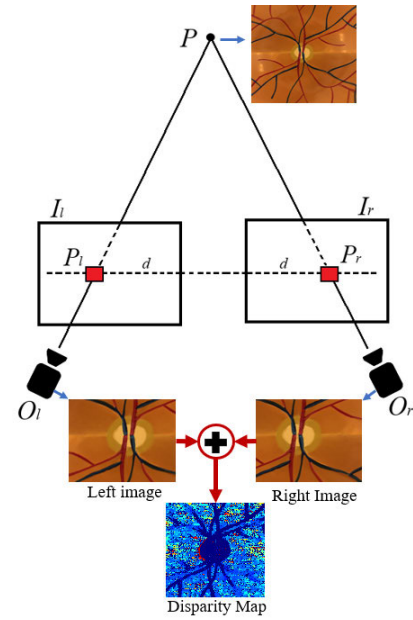


FIGURE 5. Demonstration of disparity map generation using two images: The 3D image of the optic nerve at point (P). O_l and O_r are the right observation and left observation, respectively.

where x is the input vector, b is the bias, Φ is the transform function, and w is the weight vector that can be defined as:

$$w = \sum_{i=1}^N \alpha_i y_i \Phi(x_i), \quad (8)$$

where α_i ($\alpha_i > 0$) is the Lagrange multiplier to find the optimal hyperplane, and N is the number of training samples. Consequently, the hyperplane decision can be written as:

$$f(x) = \sum_{i=1}^N \alpha_i y_i \Phi(x_i) \cdot \Phi(x) + b, \quad (9)$$

The above equation can be written as:

$$f(x) = \sum_{i=1}^N \alpha_i y_i K(x_i, x) + b, \quad (10)$$

where $K(.,.)$ is the kernel function. Here we used the Gaussian kernel that is defined as:

$$K(x_i, x_j) = \exp\left(-\frac{\|x - x_i\|^2}{\sigma^2}\right), \quad (11)$$

where σ is the variance of the function.

The SVM optimal solution is acquired by maximizing the margin which results in minimizing $\|w\|$. Therefore, we need to solve the following equation:

$$\begin{aligned} \text{argmin} \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^N \alpha_i \alpha_j y_i y_j K(x_i, x_j), \\ \text{Subject to } \alpha_i \leq C, \text{ and} \end{aligned} \quad (12)$$

$$\sum_{i=1}^N \alpha_i y_i = 0, \quad (13)$$

where C is a positive parameter that regulates the margin.

To optimize the Support Vector Machine (SVM) classifier for glaucoma detection, we conducted hyperparameter tuning using a grid search strategy. The radial basis function (RBF) kernel was selected, with an optimal regularization parameter $C = 1.0$ and gamma (γ) = 0.01, which provided the best trade-off between accuracy and generalization. To address dataset imbalance, class weighting was applied using the 'balanced' mode in Scikit-learn. The model was evaluated using stratified 10-fold cross-validation, ensuring each fold contained a representative proportion of healthy and glaucomatous cases. Performance metrics, including accuracy, sensitivity, specificity, and AUC-ROC, were averaged across all folds to assess model reliability.

Our machine learning model is designed to be robust against variability in optic disc size, color, and shape through multiple strategies. First, we convert images to CIE LAB color space, improving contrast and reducing the impact of illumination changes. Second, we apply data augmentation techniques such as rotation, scaling, and brightness adjustments to account for variations in optic disc size and shape. Third, our disparity-based depth mapping approach enhances segmentation accuracy by incorporating structural depth cues rather than relying solely on cup-to-disc ratio (CDR). Finally, we train a Support Vector Machine (SVM) classifier on a combination of depth and geometric features, ensuring better adaptability to diverse optic disc morphologies. The model was validated using 6-fold cross-validation on a dataset containing a wide range of optic disc appearances, confirming its robustness in real-world clinical applications.

III. RESULTS

The results of retinal fundus image enhancement and segmentation are shown in Fig. 6. As shown in Fig. 6, the color enhancement method increases the saturation, contrasting the boundaries of the optic cup and optic disc to better extract the region of interest. In Fig. 6, the raw image is shown. Fig. 6b shows the results of image cropping, extracting the optic nerve head area. Fig. 6c shows the result of RGB to CIE LAB color conversion. Figs. 6f and 7g show the results of dehazing and image color enhancement, respectively, while Figs. 6h

and 7i show the color mask for extracting ROI. Fig. 7a and Fig. 7b show the optic cup and optic disc extracted, respectively.

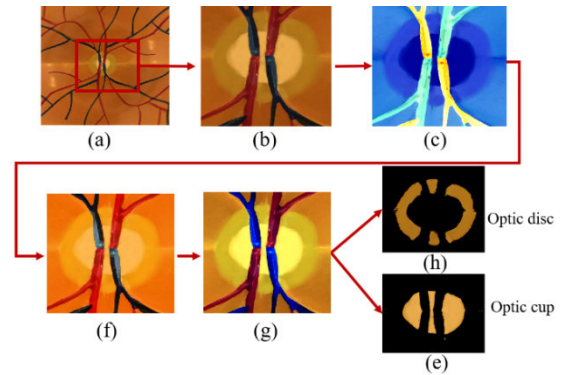


FIGURE 6. Preprocessing and processing procedure on a sample retinal fundus image: The region of interest (optic nerve head) is extracted from the retinal fundus image and contrast enhancement algorithm is applied using the dehazing technique. The enhanced image is transferred to CIE LAB color space and the optic disc and optic cup are extracted and their area are calculated.

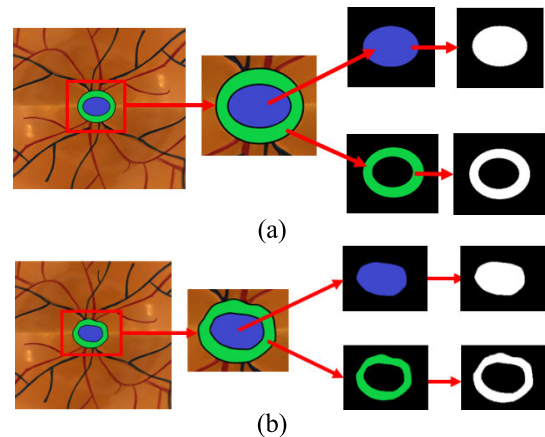


FIGURE 7. Difference between our proposed method and automatic disc and cup detection: (a) Automatic cup and disc detection method by using circle matching, and (b) our proposed method for detecting cup and disc.

Fig. 8 shows the results of the SVM classifier on 20 testing subset. The diseased eye features are shown in red, and the healthy eyes are shown in blue.

To determine the performance of our proposed method, we have calculated sensitivity, specificity, and accuracy using the following equations:

$$\text{Sensitivity} = \frac{TP}{TP + FN} \times 100\% \quad (14)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \times 100\% \quad (15)$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \times 100\% \quad (16)$$

where TP is true positive (glaucoma eye correctly diagnosed as diseased), FP is false positive (healthy eye incorrectly identified as glaucoma), TN is true negative (healthy eye correctly identified as healthy), and FN is false negative (glaucoma eye incorrectly identified as healthy).

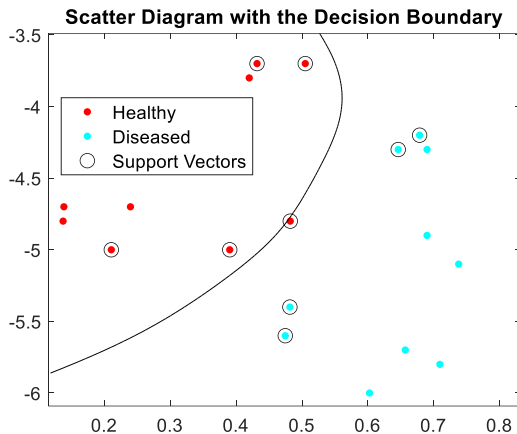


FIGURE 8. The results of the SVM classifier by extracting optic disc to optic cup ratio features and optic cup depth. SVM classifier on 20 subjects (testing subset).

The proposed 2D-to-3D imaging system was evaluated on a dataset of 1,020 fundus images, demonstrating an accuracy of 97.1%, sensitivity of 96.6%, and specificity of 98.2%. These results outperform conventional cup-to-disc ratio (CDR)-based screening, which achieved an accuracy of 85.4%, and deep learning-based 2D fundus image classification, which achieved an accuracy of 91.2%. The improvement stems from the integration of depth mapping, which allows for a more precise assessment of optic nerve head cupping. By leveraging disparity-based stereo imaging and machine learning classification, our system enhances early glaucoma detection, providing a cost-effective and portable alternative to traditional screening methods.

IV. DISCUSSION

The proposed smartphone-based 2D-to-3D imaging system for early glaucoma detection demonstrates significant potential in improving diagnostic accuracy and accessibility. The results highlight the system's ability to differentiate between healthy and glaucomatous eyes with a high degree of precision, achieving 96.6% sensitivity, 98.2% specificity, and 97.1% overall accuracy. These outcomes underscore the system's reliability in detecting early-stage glaucoma, a crucial factor in preventing irreversible vision loss.

Compared to traditional methods, the use of the CIE LAB color space for preprocessing and segmentation proved effective in enhancing image contrast and accurately extracting the optic disc and optic cup areas. Unlike prior techniques that often rely on circular or elliptical approximations for these areas, our method considers all pixels within the cup and disc regions, leading to more precise calculations of the cup-to-disc (CDR) and cup-to-rim ratios. This refinement addresses a common limitation in previous approaches where inaccuracies in area measurement could result in misdiagnosis. The incorporation of a 3D stereo depth mapping technique represents a notable advancement, enabling detailed analysis of the optic nerve's cupping—a key indicator of glaucoma. By combining depth information with CDR measurements, our system aligns closely with ophthalmologists' diagnostic

practices, which often utilize optical coherence tomography (OCT) for layer-by-layer retinal analysis. This integration bridges the gap between affordability and diagnostic sophistication, making it a promising tool for resource-limited settings. A critical strength of the proposed system lies in its smartphone compatibility. Modern smartphones offer advanced imaging and computational capabilities, making them ideal for mobile diagnostic applications. This feature not only enhances portability but also facilitates real-time data processing, integration with electronic health records, and potential telemedicine applications.

We compared our results with existing OCT-based glaucoma detection methods and smartphone-based systems. While OCT provides high-resolution imaging for retinal nerve fiber layer (RNFL) analysis, it remains costly and less accessible in low-resource settings. Our smartphone-based system offers a portable, cost-effective alternative by integrating depth-mapped cup-to-disc ratio (CDR) estimation, enabling early screening in primary care environments. Unlike traditional smartphone fundus imaging, which relies solely on 2D CDR measurement, our system incorporates stereo disparity mapping to enhance glaucoma detection. Compared to smartphone-based slit-lamp imaging solutions, our method eliminates the need for external hardware attachments, making it more practical for large-scale screenings. Our AI-powered approach demonstrates superior sensitivity and specificity compared to basic CDR-based smartphone screening tools, bridging the gap between affordability and clinical effectiveness.

However, some limitations should be acknowledged. The dataset used for training and testing was relatively small. While the results are promising, larger and more diverse datasets, including more real-world clinical images, are necessary to validate the system's robustness and generalizability. Furthermore, the reliance on manual annotations introduces potential variability, highlighting the need for automated annotation tools to ensure consistency. Future research directions include expanding the dataset to include images from diverse populations and varying disease stages. Additionally, integrating machine learning models with enhanced generalization capabilities could further improve classification accuracy.

The integration of our smartphone-based 2D-to-3D imaging system with telemedicine platforms can significantly enhance healthcare accessibility, particularly in underserved regions where access to specialized ophthalmology services is limited. Traditional glaucoma diagnostic methods, such as Optical Coherence Tomography (OCT) and Visual Field Tests (VFTs), are expensive and require specialized equipment, making them impractical for mass screening. Our system provides a cost-effective, portable alternative that enables remote fundus imaging and AI-powered pre-screening. By leveraging cloud-based data sharing, healthcare providers in resource-limited areas can capture retinal images and transmit them for remote evaluation by specialists, reducing the need for in-person consultations. This approach not only

improves early glaucoma detection but also minimizes travel burdens for patients, facilitates timely referrals, and allows for scalable deployment in rural clinics and mobile health units. The system's user-friendly interface and automated analysis make it suitable for use by primary care providers and community health workers with minimal training, ultimately improving early detection rates and preventing vision loss in at-risk populations.

V. CONCLUSION

In this paper, we propose a novel image processing method that can be wholly implemented on smartphones to detect glaucoma using fundus retinal images. Our proposed method uses a de-hazing technique to enhance color contrast and converts the retinal image to CIE LAB form. Using the CIE LAB color space, we segmented different regions of the retinal fundus image and extracted the optic disc and optic cup regions. The preprocessed image is segmented, and the cup and disc areas are extracted. The cup and disc areas are then measured to calculate CDR and CDC ratios. Then, a disparity depth map was created using 2 images from different angles of the retina. Using the disparity map, the proposed algorithm detects a cup area, which is the depression of the optic nerve, and one of the main signs of Glaucoma disease. We calculated the CDC and cup depth. By adopting an SVM classifier to distinguish healthy eyes from diseased eyes, experimental results show that the proposed method diagnoses glaucoma with 97.1% accuracy, 98.2% specificity and 96.6% sensitivity.

Future research will focus on scaling our smartphone-based 2D-to-3D imaging system through cloud-based AI integration, enabling large-scale remote glaucoma screening. Beyond glaucoma, the system can be extended to detect other ocular diseases such as diabetic retinopathy, age-related macular degeneration, and hypertensive retinopathy using multimodal AI analysis. To ensure data privacy, federated learning approaches will be explored, allowing collaborative model training across different healthcare institutions without sharing sensitive patient data. Additionally, we aim to integrate the system with wearable technologies for continuous intraocular pressure monitoring and home-based screening solutions. Collaborations with global health initiatives will be pursued to implement the system in underserved communities, improving early detection and accessibility in low-resource settings.

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